Logistic Regression and Classification

Section 1: The Odds Ratio and Logit Formulation

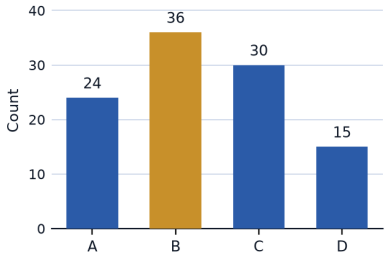
1 If a classifier predicts probability $p = 0.80$ for disease occurrence, what are the odds of having the disease?

a. 4.0

b. 0.25

c. 0.80

d. 5.0



2 Compute the log-odds (logit) value when probability $p = 0.50$.

a. 0.0

b. 1.0

c. 0.5

d. -1.0

3 If a logistic model has equation $\ln\left(\frac{p}{1-p}\right) = 2.0$, find the predicted probability p.

a. $\frac{e^2}{1+e^2} \approx 0.88$

b. $\frac{1}{2} = 0.50$

c. $\frac{2}{3} \approx 0.67$

d. 0.95

Section 2: Binary Cross-Entropy Loss

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4 Calculate the binary cross-entropy loss $L = -[y \ln \hat{y} + (1-y)\ln(1-\hat{y})]$ for a sample where true label $y = 1$ and predicted probability $\hat{y} = 0.90$.

a. $-\ln(0.90) \approx 0.105$

b. $-\ln(0.10) \approx 2.30$

c. 0.90

d. 0.0

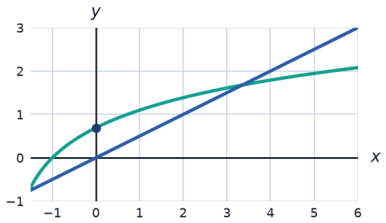
5 What is the derivative of the binary cross-entropy loss with respect to the pre-activation logit z, where $\hat{y} = \sigma(z)$?

a. $\hat{y} - y$

b. $\hat{y}(1-\hat{y})$

c. $y - \hat{y}$

d. $\frac{\hat{y}-y}{\hat{y}(1-\hat{y})}$



6 What happens to the cross-entropy loss as $\hat{y} \to 0$ when true label is $y = 1$?

a. It approaches $+\infty$, heavily penalizing confident wrong predictions

b. It approaches zero

c. It oscillates around 1

d. It equals the pre-activation logit

Section 3: The Confusion Matrix and Core Metrics

7 A diagnostic model reports $TP = 70, TN = 20, FP = 5, FN = 5$. Compute the Accuracy.

a. 90%

b. 85%

c. 93.3%

d. 75%

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8 Using the same counts ($TP = 70, FP = 5, FN = 5$), calculate the model's Precision.

a. $\frac{70}{75} \approx 93.3\%$

b. $\frac{70}{70+5} = 90\%$

c. $\frac{70}{100} = 70\%$

d. $\frac{75}{100} = 75\%$

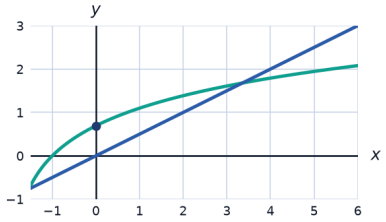
9 Calculate the Recall (Sensitivity) for the model above.

a. $\frac{70}{75} \approx 93.3\%$

b. $\frac{70}{90} \approx 77.8\%$

c. $\frac{5}{75} \approx 6.7\%$

d. 85%



Section 4: The F_1-Score, Thresholding and ROC-AUC

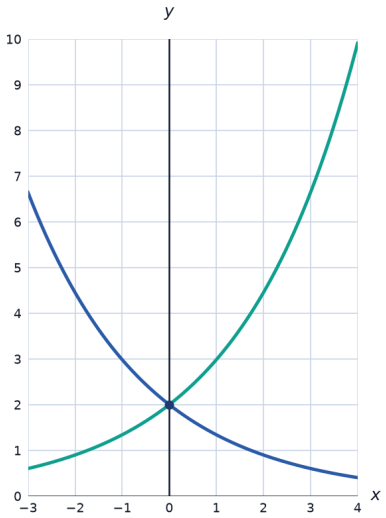
10 Compute the F_1-score for a model having Precision $= 0.80$ and Recall $= 0.60$.

a. $\frac{2(0.8)(0.6)}{0.8+0.6} \approx 0.686$

b. 0.700

c. 0.480

d. 0.750



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11 If you lower the classification probability threshold from 0.5 to 0.2, what is the typical effect on Precision and Recall?

a. Recall increases (fewer false negatives), while Precision typically decreases (more false alarms)

b. Precision increases while Recall decreases

c. Both metrics increase simultaneously

d. Neither metric changes

12 What does an Area Under the ROC Curve (ROC-AUC) of 0.50 indicate about a classifier?

a. The model has no discriminative ability (equivalent to random guessing)

b. The model is a perfect classifier

c. The model achieves 50% precision at all thresholds

d. The model is inverted

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